Worksheet & Answer Key

Carbon compounds exhibit unique chemical properties due to the ability of carbon atoms to form strong bonds with themselves and other atoms. This property, known as _____, allows for the creation of complex molecular structures. In the case of oxidation, carbon compounds can react with oxygen to form _____ compounds, releasing energy in the process.

Word Bank:

polymerize, reactive, energetic, complexity, strongly, covalently, oxidized, release

Fill in the blanks:

1. Carbon compounds exhibit unique chemical properties due to the ability of carbon atoms to form _____ bonds with themselves and other atoms.

2. The property, known as _____, allows for the creation of complex molecular structures.

3. This property enables carbon compounds to _____ with oxygen to form oxidized compounds.

4. Oxidation can result in the release of _____ energy in carbon compounds.

5. In the presence of oxygen, carbon compounds can _____ into oxidized forms.

6. The strong bonds between carbon atoms enable them to _____ and form complex molecules.

7. The ability of carbon atoms to form strong bonds with other atoms is a key aspect of their _____ nature.

8. Carbon compounds can be _____ by oxidation reactions, releasing energy in the process.

9. This unique property allows carbon compounds to be highly _____ and reactive.

■ Answer Key (for teachers only):

1. covalently
2. polymerize
3. energetic
4. complexity
5. strongly
6. covalently
7. oxidized
8. release